Module-1 Introduction to Knowledge Management

History, From Physical Assets to Knowledge Assets, Organizational Perspective, The Concept Analysis Technique and The Types of Knowledge

KM: History

Knowledge Management Milestones

Year	Entity	Event
1980	DEC, CMU ★	XCON Expert System
1986	Dr. K. Wiig	Coined KM concept at UN
1989	Consulting firms	Start internal KM projects
1991	HBR article	Nonaka & Takeuchi
1993	Dr. K. Wiig	First KM book published
1994	KM Network	First KM conference
Mid-1990s	Consulting firms	Start offering KM services
Late 1990s	Key vertical industries	Implement KM and start seeing benefits
2000–2003	Academia	KM courses/programs in universities with KM texts

★**DEC- Digital Equipment Corporation of Carnegie Mellon University (CMU)**
More about KM History can be studied from the provided study materials.

KM: From Physical Assets to Knowledge Assets

- **Knowledge has become increasingly more valuable than the more traditional physical or tangible assets.** Take an example of an **Airline company**
  - **Traditionally**, **company's most valuable asset** was the **physical inventory of airplanes**.
  - **Today**, the airlines' **greatest asset is the reservation system** that enables the airline not only to manage the logistics of its passenger reservations but also to implement a seat "**yield management system**."
  - The **yield management system** refers to an **optimization program** that is used to ensure that **maximum revenue** is generated from each seat sold—even if each and every seat carries a distinct price.
- This is an example of ***Intellectual assets***, which generally refers to an organization's **recorded information, and human talent.**
  - But such information is typically either inefficiently warehoused or simply lost, especially in large, physically dispersed organizations.

KM: From Physical Assets to Knowledge Assets (cont...)

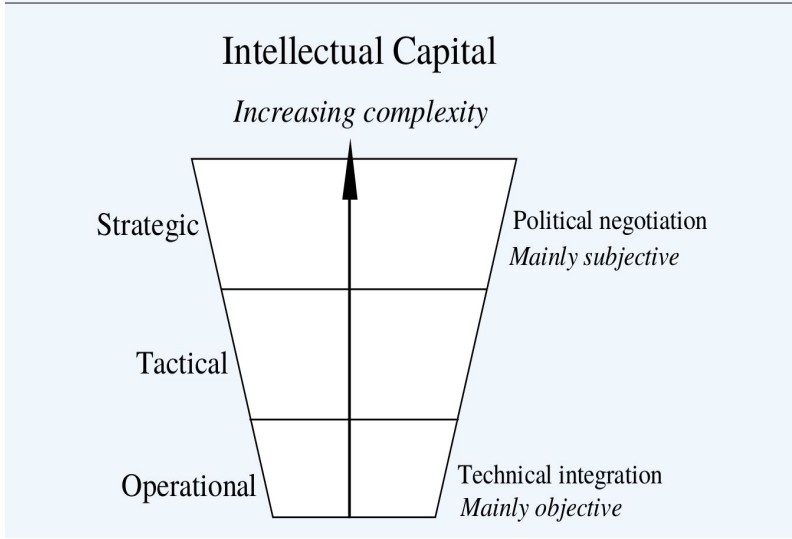
Modern day approach is that Knowledge never loses its value.

- It is no longer advisable to simply discard items that are "past their due date." Instead, **content analysis** and a **cost-benefit analysis** are needed to manage each piece of valuable knowledge in the best possible way.
- Intellectual capital is often made visible by the **difference between the book value and the market value** of an organization.
- Intellectual assets are represented by **the sum total of what employees of the organization know and what they know how to do**.
- The value of these knowledge assets is at least equal to the cost of recreating this knowledge.

KM: From Physical Assets to Knowledge Assets (cont…)

- Some examples of intellectual capital include:
  - **Competence**—the skills necessary to achieve a certain (high) level of performance.
  - **Capability**—strategic skills necessary to integrate and apply competencies.
  - **Technologies**—tools and methods required to produce certain physical results.
- The **accounting or measuring of these new forms of assets still has considerable difficulty**
- **Skandia** (a private financial services company from the Stockholm, Sweden) was the first organization to report intellectual capital as part of its yearly financial report.
- Intellectual assets may be found at the strategic, tactical, and operational levels of an organization.

THREE LEVELS OF INTELLECTUAL CAPITAL



Organizational Perspectives on Knowledge Management

- Wiig (1993) considers knowledge management in organizations from three perspectives, each with different horizons and purposes:
  - **Business Perspective**—focusing on **why, where, and to what extent the organization must invest in or exploit knowledge**. Strategies, products and services, alliances, acquisitions, or divestments should be considered from knowledge-related points of view.
  - **Management Perspective**—focusing on **determining, organizing, directing, facilitating, and monitoring** knowledge-related practices and activities required to achieve the desired business strategies and objectives.
  - **Hands-on Perspective**—focusing on **applying the expertise** to conduct explicit knowledge-related work and tasks.
- The **business perspective** easily maps onto the **strategic nature** of knowledge management, the **management perspective** is parallel to the **tactical layer**, and the **hands-on perspective** may be equated with the **operational level**. (see the figure in previous slide)

The Concept Analysis Technique

- Concept analysis is an **established technique** used in the social sciences, such as philosophy and education, in order **to derive a "formula"** that in turn can be **used to generate definitions and descriptive phrases for highly complex terms**.
- The concept analysis approach rests on obtaining consensus on three major dimensions of a given concept:
  - A list of **key attributes** that must be present in the definition, vision, or mission statement.
  - A list of **illustrative examples.**
  - A list of **illustrative nonexamples**.
- This approach is particularly **useful in tackling multidisciplinary domains** such as intellectual capital, **for clear criteria** can be developed to enable sorting into categories such as **knowledge versus information**, **document management versus knowledge management**, and **tangible versus intangible assets**.

The Concept Analysis Technique (cont…)

Ruggles and Holtshouse (1999) identified the following key attributes of knowledge management to define KM using the concept analysis technique:

- Generating new knowledge.
- Accessing valuable knowledge from outside sources.
- Using accessible knowledge in decision making.
- Embedding knowledge in processes, products, and/or services.
- Representing knowledge in documents, databases, and software.
- Facilitating knowledge growth through culture and incentives.
- Transferring existing knowledge into other parts of the organization.
- Measuring the value of knowledge assets and/or impact of knowledge management.

The Concept Analysis Technique (cont…)

- It is **highly recommended that each organization undertake the concept analysis exercise to clarify its understanding of what KM means in its own organization's context.**
- The best way to do so would be to **work as a group**, enabling them to **achieve a shared understanding at the same time** that they develop a clearer conceptualization of the KM concept. **Each participant** can take a turn **contributing one good example of what KM is and another example of what KM is not**.
- The **entire group can then discuss this example & non-example pair** in order to identify one (or several) key KM attributes.
- Once the group members feel they have covered as much ground as they are likely to, the key attributes can be summarized in the form of a **KM concept "formula" such as:** "In our organization, knowledge management must include the following: both tacit and explicit knowledge; a framework to measure the value of knowledge assets; a process for managing knowledge assets. . . . and so on."

The Two Major Types of Knowledge: Tacit vs Explicit

Tacit knowledge

- It is **difficult to articulate** and also **difficult to put into words, text, or drawings**.
- Tacit knowledge tends to reside "**within the heads of knowers**,".
- Highly skilled, experienced, and expert individuals may find it **harder to articulate their know-how because of the "tacitness".**

Explicit knowledge

- It represents content that has been **captured in some tangible form such as words, audio recordings, or images**.
- Explicit knowledge is usually contained **within tangible or concrete media**.
- Novices, on the other hand, are more apt to easily verbalize what they are attempting to do because they are **typically following a manual or how-to process in the form of explicit knowledge**.

The Two Major Types of Knowledge: Tacit vs Explicit (cont...)

- The **same content** may be **explicit for one person** and **tacit for another**.
- This can be better understood by the **paradox of knowledge of something versus knowledge of how to do something**.
- Another perspective is that **explicit knowledge tends to represent the final end product**, whereas **tacit knowledge is the know-how** or **all of the processes** that were required in order **to produce that final product**.

The Two Major Types of Knowledge: Tacit vs Explicit (cont...)

- A popular **misconception** is that **KM focuses on rendering whatever is tacit into more explicit or tangible forms**, and then **storing or archiving these forms somewhere**, usually in some form of **intranet or knowledge portal**.
- In fact, knowledge management is **broader** and **includes leveraging the value of the organizational knowledge** and **know-how** that **accumulates over time**.
- This approach is **much more holistic and user-centered** and begins **not with an audit of existing documents** but with a **needs analysis** to better **understand how improved knowledge sharing may benefit specific individuals, groups, and the organization as a whole.**
- **Successful knowledge-sharing examples are gathered and documented in the form of lessons learned and best practices, and these then form the kernel of organizational stories.**